A Generative AI Framework for Contractor Cost Analysis

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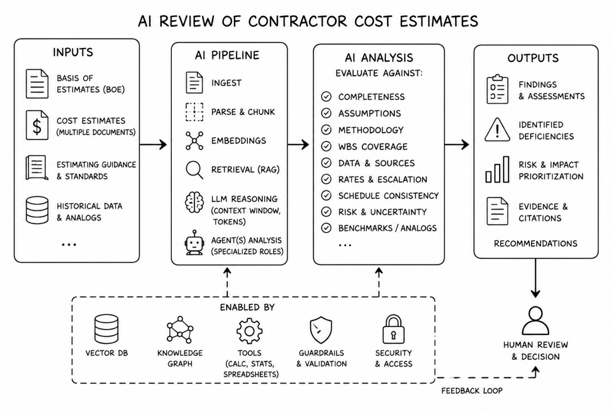
July 29, 2026

Mission Statement

What information must be provided to a generative AI system to enable credible evaluation of cost estimates?

Schematic


AI REVIEW OF CONTRACTOR COST ESTIMATES
INPUTS
BASIS OF ESTIMATES (BOE)
COST ESTIMATES (MULTIPLE DOCUMENTS)
ESTIMATING GUIDANCE & STANDARDS
HISTORICAL DATA & ANALOGS
...
AI PIPELINE
INGEST
PARSE & CHUNK
EMBEDDINGS
RETRIEVAL (RAG)
LLM REASONING (CONTEXT WINDOW, TOKENS)
AGENT(S) ANALYSIS (SPECIALIZED ROLES)
AI ANALYSIS
EVALUATE AGAINST:
COMPLETENESS
ASSUMPTIONS
METHODOLOGY
WBS COVERAGE
DATA & SOURCES
RATES & ESCALATION
SCHEDULE CONSISTENCY
RISK & UNCERTAINTY
BENCHMARKS / ANALOGS
...
OUTPUTS
FINDINGS & ASSESSMENTS
IDENTIFIED DEFICIENCIES
RISK & IMPACT PRIORITIZATION
EVIDENCE & CITATIONS
RECOMMENDATIONS
ENABLED BY
VECTOR DB
KNOWLEDGE GRAPH
TOOLS (CALC, STATS, SPREADSHEETS)
GUARDRAILS & VALIDATION
SECURITY & ACCESS
HUMAN REVIEW & DECISION
FEEDBACK LOOP

Presentation Overview I

Presentation Overview II

- GAO
- Commerce
- Contractors

Input to AI

1. Contractor estimates
2. Historical estimates
3. Cost-estimating handbook
4. Program documentation
5. Acquisition regulations

AI Pipeline

1. Document ingestion
2. Parsing
3. Chunking
4. Embedding
5. Retrieval
6. Prompt construction
7. LLM reasoning
8. Report generation

AI Components

1. Retrieval pipeline (RAG)
2. Agent(s)
3. Prompt templates
4. Token budgeting
5. Context window management
6. Human review

Outputs

1. Findings
2. Missing assumptions
3. Inconsistencies
4. Confidence
5. Evidence citations

Where Does This Research Stand?

1. Tasking
2. Online Library
3. Survey of AI Landscape

Collection of Tasking Documents

emails, memos, phone calls, ...

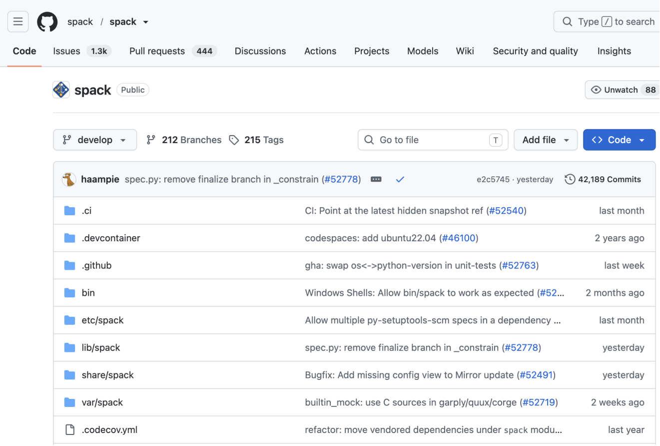
Primary Players

1. Envision by Palantir
2. Gemini
3. Gemma
4. GenAI.mil

GitLab Repo

Best Practice: Organize data in GitLab

GitLab Repo Example I



GitLab Repo Example II

README | Code of conduct | Contributing | Apache-2.0 license | MIT license | Security

Spack

ci passing | Bootstrapping passing | Containers passing | docs passing | codecov 84% | slack 4207 | matrix 50 users

Getting Started • Config • Community • Contributing • Packaging Guide • Packages

Spack is a multi-platform package manager that builds and installs multiple versions and configurations of software. It works on Linux, macOS, Windows, and many supercomputers. Spack is non-destructive: installing a new version of a package does not break existing installations, so many configurations of the same package can coexist.

Spack offers a simple "spec" syntax that allows users to specify versions and configuration options. Package files are written in pure Python, and specs allow package authors to write a single script for many different builds of the same package. With Spack, you can build your software *all* the ways you want to.

See the Feature Overview for examples and highlights.

Plan B: Schema

```
p:\waynes-world/
 ├── components
 ├── inputs
 ├── outputs
 ├── pipeline
 └── tasking
```

Directory vs. Repository

Directory	Repository
Siloed, local only	Shared, collaborative
No history	Full audit trail
No branching	Unlimited branches
Manual backups	Automated remotes
No access control	Fine-grained permissions
No tools	Full dev toolchain
Risk of loss	Backup by design

A plain directory is just storage. A repository is a living system for building software and cataloging results.

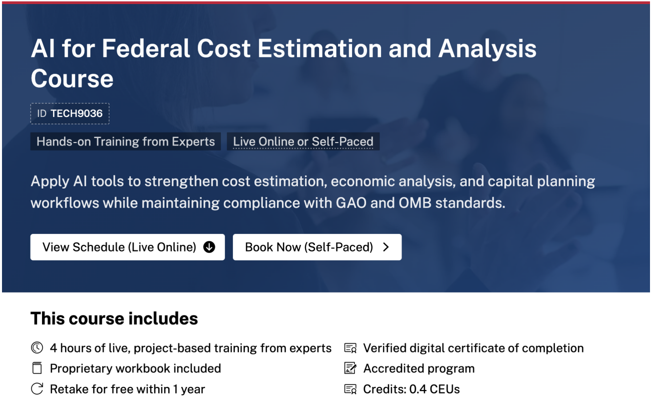
Where Does This Research Belong?

Where Does This Research Belong?

- Can machine learning produce better estimates?
- Can AI automate portions of the estimating workflow?
- Can generative AI help an estimator prepare a basis of estimate?
- Can AI retrieve analogous historical programs or identify cost drivers?

What AI architecture and domain knowledge are required to evaluate a collection of cost estimates credibly?

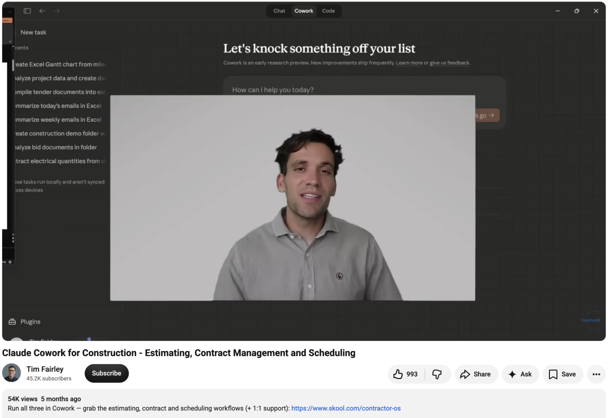
AI for Federal Cost Estimation and Analysis



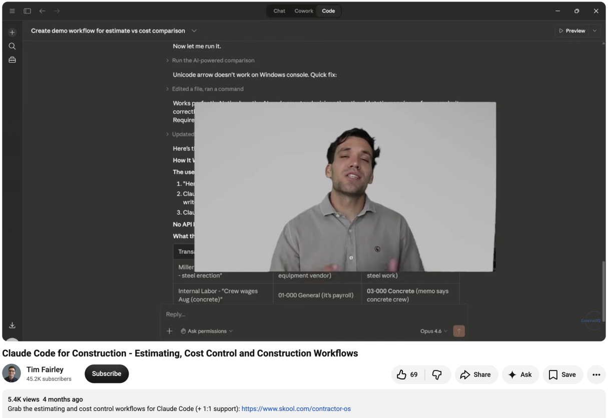
AI for Federal Cost Estimation and Analysis

*This half-day course teaches federal cost professionals how to use **AI as a practical tool** for developing, **reviewing**, and communicating **cost estimates** and economic analyses. Participants will learn to **apply AI** to lifecycle cost modeling, cost-benefit and cost-effectiveness analysis, **independent cost estimate review**, and capital planning — while maintaining compliance with GAO Cost Estimating and Assessment Guide standards and OMB requirements.*

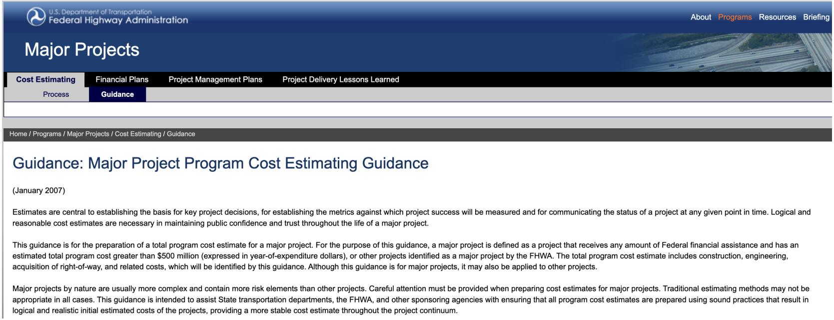
Claude Cowork for Construction - Estimating



Claude Cowork for Construction - Estimating, Cost Control



Major Project Program Cost Estimating Guidance



U.S. Department of Transportation
Federal Highway Administration

About Programs Resources Briefing

Major Projects

Cost Estimating | Financial Plans | Project Management Plans | Project Delivery Lessons Learned

Process | Guidance

Home / Programs / Major Projects / Cost Estimating / Guidance

Guidance: Major Project Program Cost Estimating Guidance

(January 2007)

Estimates are central to establishing the basis for key project decisions, for establishing the metrics against which project success will be measured and for communicating the status of a project at any given point in time. Logical and reasonable cost estimates are necessary in maintaining public confidence and trust throughout the life of a major project.

This guidance is for the preparation of a total program cost estimate for a major project. For the purpose of this guidance, a major project is defined as a project that receives any amount of Federal financial assistance and has an estimated total program cost greater than $500 million (expressed in year-of-expenditure dollars), or other projects identified as a major project by the FHWA. The total program cost estimate includes construction, engineering, acquisition of right-of-way, and related costs, which will be identified by this guidance. Although this guidance is for major projects, it may also be applied to other projects.

Major projects by nature are usually more complex and contain more risk elements than other projects. Careful attention must be provided when preparing cost estimates for major projects. Traditional estimating methods may not be appropriate in all cases. This guidance is intended to assist State transportation departments, the FHWA, and other sponsoring agencies with ensuring that all program cost estimates are prepared using sound practices that result in logical and realistic initial estimated costs of the projects, providing a more stable cost estimate throughout the project continuum.

Major Project Program Cost Estimating Guidance

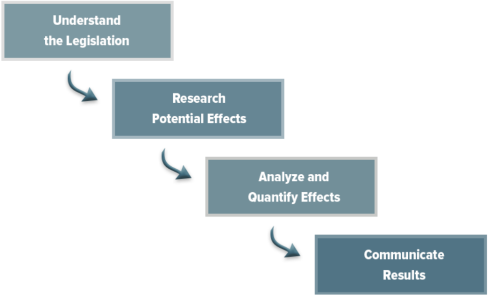
Table of Contents

Congressional Budget Office

April 2023

CBO Describes Its Cost-Estimating Process

CBO's Four-Step Process for Preparing Cost Estimates



Report's Home Page
Related Publications
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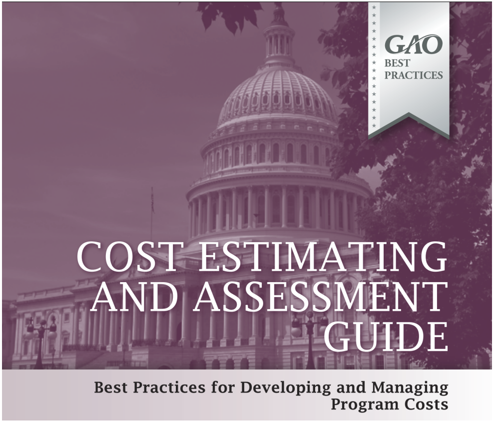
Government Accountability Office

GAO COST ESTIMATING AND ASSESSMENT GUIDE

Government Accountability Office: Best Practices



Department of Commerce

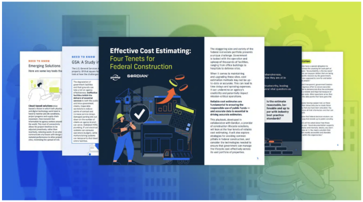
Department of Commerce

Cost Estimating Guide

Version 1.1

Federal Pricing Group

Gordian



Effective Federal Cost Estimation: Four Tenets of Reliable Estimates

October 8, 2024

By Gordian

The staggering size and variety of the **federal** real estate portfolio presents a unique challenge. Government is tasked with the operation and upkeep of thousands of facilities, ranging from office buildings to hospitals to defense sites.

References I

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